SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Constituent of Symbiosis International (Deemed University), Pune

Project Title:

Longevity Lens: Decoding Life Expectancy Trends Worldwide

Course: AIML - 2025-29 Sem II -TIL: Tinker and IDEA Lab

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1. Introduction

Longevity Lens: Decoding Life Expectancy Trends Worldwide is a data visualization project developed using Power BI to analyse and understand global patterns in life expectancy. The project focuses on exploring how various factors such as GDP, mortality rates, health indicators, and disease prevalence influence life expectancy across different countries. By using a structured dataset and applying data preprocessing techniques, the project aims to transform raw data into meaningful insights that can support better understanding of global health trends.

The dashboard is designed to be interactive and user-friendly, allowing users to explore data through multiple visualizations such as KPI cards, charts, maps, and slicers. It enables comparison between developed and developing countries, analysis of trends over time, and evaluation of key health indicators. Overall, this project helps in identifying important relationships between economic and health factors, providing a clear and comprehensive view of life expectancy trends worldwide.

2. Problem Statement

Understanding the factors that influence life expectancy across different countries is a complex task due to the presence of large, diverse, and unstructured datasets. Key variables such as GDP, mortality rates, disease prevalence, and health indicators vary significantly across regions, making it difficult to identify clear patterns and relationships using raw data alone. Without proper analysis and visualization, it becomes challenging to derive meaningful insights that can help in comparing developed and developing countries or tracking changes over time.

This project aims to address this challenge by developing an interactive Power BI dashboard that analyzes global life expectancy data. The objective is to preprocess and visualize the data in a structured manner, enabling users to explore trends, compare countries, and understand how different health and economic factors impact life expectancy.

3. Objectives

The objective of this project is to build a user-friendly, interactive and visually appealing dashboard with the help of power BI to analyse life expectancy data. Power BI is applied in this project to transform raw data into structured visual reports. The key metrics mentioned in the project includes life expectancy, GDP, population, adult mortality, infant deaths and various health indicators across different countries and years. We also aimed to examine how socio-economic and health related issues are in correlation and can affect the overall life expectancy. The dashboard allows us to visualize the interconnections between the data and how data changes with time. We can distinctly understand the various trends of different countries and understand the comparison between them. To design the data visually effectively and easy to understand we can implement multiple visual elements such as bar graphs, pie charts, column charts/vertical bars, tree maps, waterfall charts and tables. Moreover, power BI also includes slicers and filters to enable dynamic interaction with data. Besides the above-mentioned functions this project is also involved in the process of data preprocessing using Power Query, including correcting data types, renaming columns, and handling missing values. Overall, the objective was to derive meaningful insights from real-world data and present them in a clear and interactive manner.

4. Dataset Description

The dataset for this project was acquired from Kaggle and its data demonstrates the trends of life expectancy across multiple countries over several years. This data contains over 2900 records and includes various pieces of information that helps us to effectively form an analysis.
Dataset link- https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who

Here both numerical and categorical types of data are available. The most significant columns of the dataset include the country, its GDP, life expectancy, literacy rate, BMI (Body Mass Index), total no of population, alcohol consumption, year (Date\Time), Adult Mortality, Health Expenditure, Infant Deaths, Status (Developed/Developing), and Immunization-related factors.
Furthermore, we can differentiate between the developed and developing countries while looking at this data and comprehend the contrast between different socio-economic groups.
Here, we can observe the various trends, patterns and alternations on life expectancy over the years and closely examine the interconnections between factors affecting life expectancy.
We can investigate if people live longer when they have healthcare, more money and better living conditions. While preparing the data for the dashboard we encountered missing values which then were replaced using Power BI's Power Query Editor. Replacing missing values null with zeros is an important step to maintain consistency of the data. Without this step we cannot perform proper analysis of the data. To further improve the quality of data columns names were standardized and data types were also corrected.
This data is really useful because we can see how different things like money, education and health affect how long people live. We can use this data to learn about health trends around the world and how rich and poor countries are different. We can also see what is important for people to live longer. The data is a base for making interactive pictures and coming to conclusions that are based on facts. We can learn a lot from the life expectancy data. It can help us understand what makes people live longer. The life expectancy data is a tool for people who want to know more about how to stay healthy and live longer.

5. <u>Data Preprocessing</u>

Data preprocessing is a crucial step in any data analysis or machine learning workflow, where raw data is cleaned and transformed into a structured and usable format. It involves tasks such as handling missing values, correcting data types, removing inconsistencies, and preparing the dataset for further analysis or visualization. We identified the need for data preprocessing by examining the dataset using the Pandas library in Jupyter Notebook. Specifically, functions like

- df.info()
- df.isnull().sum()

allowed us to detect columns with missing (null) values. Through this analysis, we were able to identify inconsistencies such as incorrect data types and the presence of null values, which indicated that preprocessing was necessary before proceeding further.

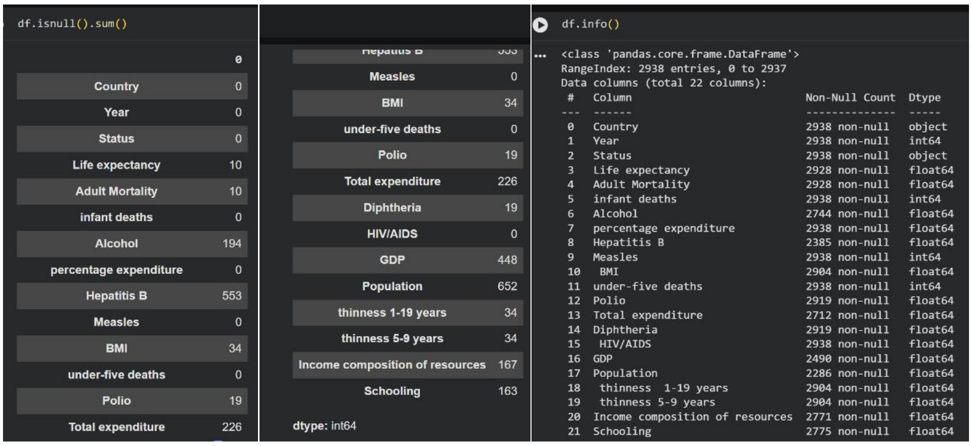


Figure: Output from Google Colab demonstrating the use of the Pandas library functions df.info() and df.isnull().sum() to inspect the dataset structure, identify data types, and detect missing (null) values in each column.

To ensure data quality and consistency for analysis, several preprocessing steps were performed on the dataset:

a) <u>Column Renaming</u>

Certain column names were modified to improve readability and maintain consistency:

- life expectancy → Life Expectancy
- HIV/AIDS → HIV AIDS
- Income composition of resources → Income Composition of Resources

Column Renaming in Power BI

Column renaming in Power BI was performed using the Power Query Editor, which is accessed through the Transform Data option. Upon selecting this option, a separate window (Power Query Editor) opens, providing a dedicated environment for data transformation and cleaning. Within this interface, each column was individually selected and renamed to more meaningful and standardized names. This step ensures better readability, consistency, and ease of understanding while building reports and visualizations. Renaming columns at this stage also helps maintain a structured data model, making it more efficient to reference fields during dashboard creation and analysis.

b) <u>Handling Missing Values</u>

The dataset contained missing (null) values in multiple columns. These were handled by replacing null values with 0 to maintain uniformity and avoid computational errors during analysis.

The following columns were treated:

- Schooling
- Income Composition of Resources
- Thinness 5–9 years
- Thinness 10–19 years
- Population
- GDP
- Diphtheria
- Total Expenditure
- Polio
- BMI
- Hepatitis B
- Alcohol
- Life Expectancy
- Adult Mortality

Handling Missing Values using Replace Values in Power BI

In the Power Query Editor, select the column containing null values, right-click on it, and choose Replace Values. In the dialog box, enter null in the Value to Find field and 0 in the Replace With field, then click OK.

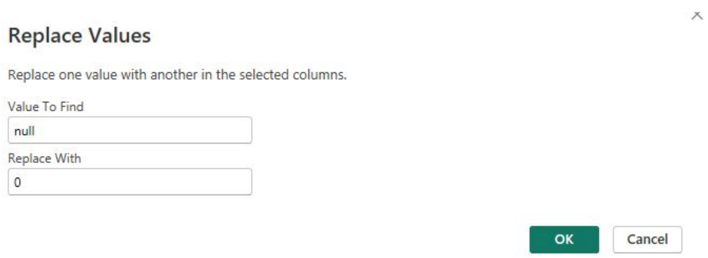


Figure: Power Query Editor in Power BI showing the use of the Replace Values option to replace null values with 0 in the selected column.

c) <u>Data Type Correction</u>

To ensure proper data interpretation and analysis, incorrect data types were corrected:

- Year: Converted from int64 to Date format
- Life Expectancy: Converted from float64 to int64

Data Type Correction in Power BI

In the Power Query Editor, select the desired column, right-click on it, and choose the Change Type option. From the list of available data types (such as Text, Whole Number, Decimal Number, Date, etc.), select the appropriate type based on the data.

These preprocessing steps helped in improving the dataset's structure, handling inconsistencies, and preparing it for accurate analysis and visualization.

6. Dashboard Features

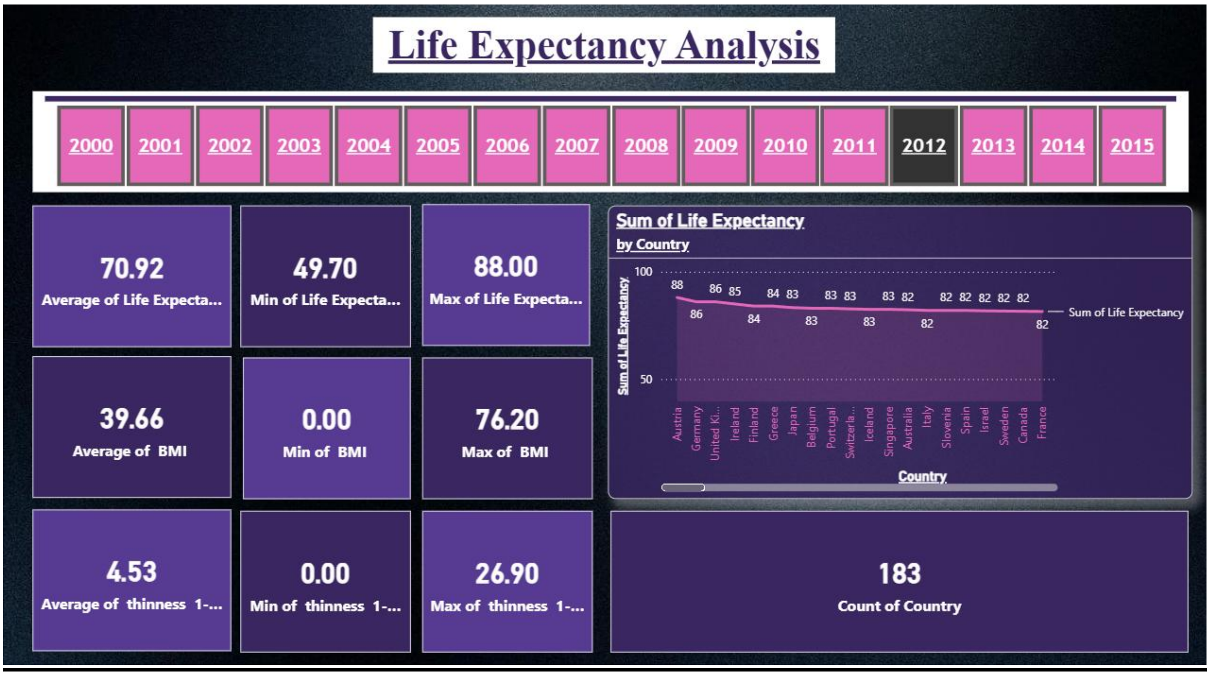
A dashboard in Power BI is an interactive visual interface that presents data in a clear and meaningful way using different types of visuals. It helps users quickly understand patterns, trends, and insights by combining multiple elements such as charts, KPI cards, slicers, and tables in a single view. These features make the dashboard dynamic and user-friendly, allowing users to filter data, compare values, and analyse relationships between different variables. Different types of dashboard components include KPI cards for summary values, charts for comparisons and trends, maps for geographical analysis, slicers for filtering, and tables for detailed data representation.

Power BI provides two main panels that help in creating and customizing these effects. The Visualizations Pane is used to select different types of visuals (such as bar charts, line charts, cards, etc.) and to assign data fields like axis, values, legend, and filters. The Fields Pane contains all the dataset columns, which can be dragged and dropped into visuals to build the dashboard. Additionally, the Format Pane (paint roller icon) is used to customize the appearance of visuals, including colours, titles, labels, and layout.

Features Used in the Dashboard

- KPI Cards (for summary metrics like average, min, max)
- Bar and Column Charts (for comparison between countries)
- Line and Combo Charts (for trend and relationship analysis)
- Donut/Pie Charts (for distribution analysis)
- Slicers (for filtering by Year and Country)
- Map Visualization (for geographical representation)
- Tables/Matrix (for structured data view)
- Legends (to differentiate categories like Developed/Developing)
- Data Labels and Titles (for better readability)
- Colour Formatting and Shapes (for design and layout)

DASHBOARD PAGE 1: LIFE EXPECTANCY ANALYSIS



a) **KPI Cards**

KPI cards are used to present key summary statistics such as:

- Average, Minimum, and Maximum of Life Expectancy
- Average, Minimum, and Maximum of BMI
- Average, Minimum, and Maximum of Thinness
- Count of Countries

These cards provide a quick overview of the dataset and help in understanding overall trends instantly.

How it was created:

Fields like Life Expectancy, BMI, and Thinness were dragged into the Card visual, and aggregation functions such as Average, Minimum, and Maximum were applied.

b) **Year Slicer (Filter)**

A horizontal slicer displaying years (2000–2015) is used to filter the entire dashboard dynamically. It allows users to analyze data for a specific year.

How it was created:

The Year column was dragged into the Slicer visual and formatted horizontally.

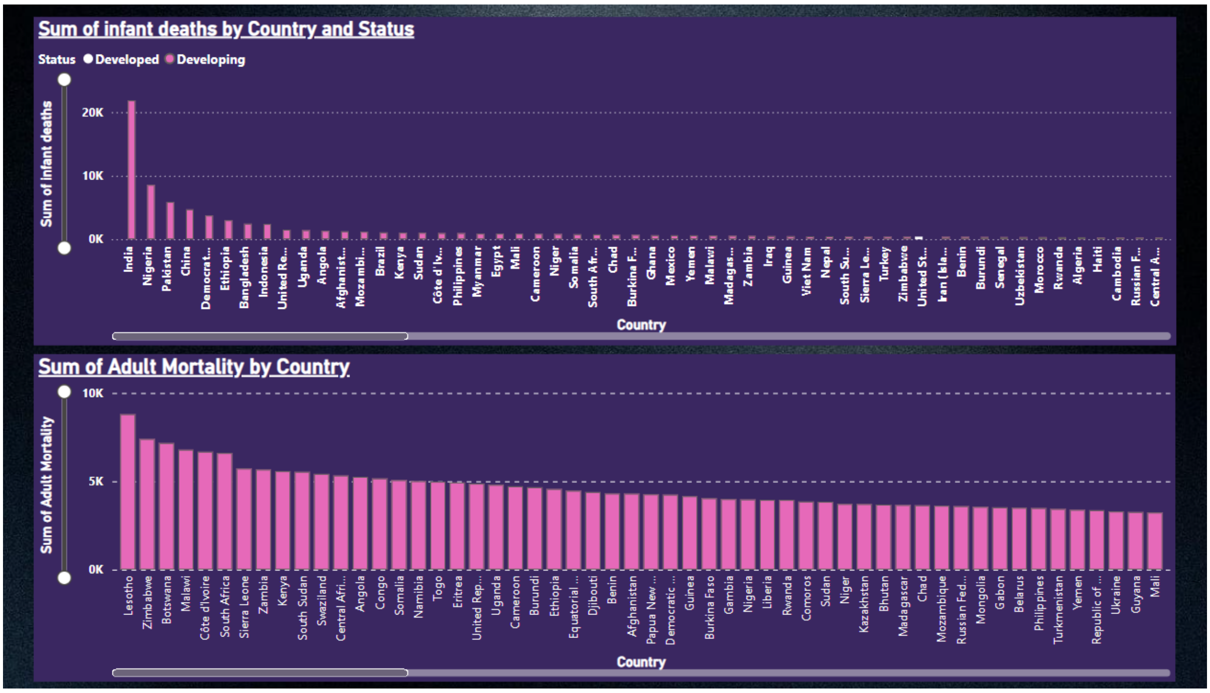
c) **Bar Chart – Life Expectancy by Country**

This chart shows the sum of life expectancy across countries, enabling comparison between different countries. It highlights which countries have higher or lower life expectancy.

How it was created:

In this visualization, the Country field is placed on the Axis to represent different countries, while the Life Expectancy field is added to the Values section to display the corresponding life expectancy for each country. This helps in comparing life expectancy across multiple countries in a clear and structured manner.

DASHBOARD PAGE 2: INFANT AND ADULT MORTALITY



a) **Clustered Bar Chart – Infant Deaths by Country and Status**

This chart compares infant deaths across countries, categorized by Developed and Developing status. Shows how infant mortality differs based on country development status.

How it was created:

In this visualization, the Country field is placed on the Axis to represent different countries, while Infant Deaths is added to the Values section to show the number of infant deaths. The Status field is used in the Legend to differentiate between developed and developing countries, making it easier to compare infant mortality across different country categories.

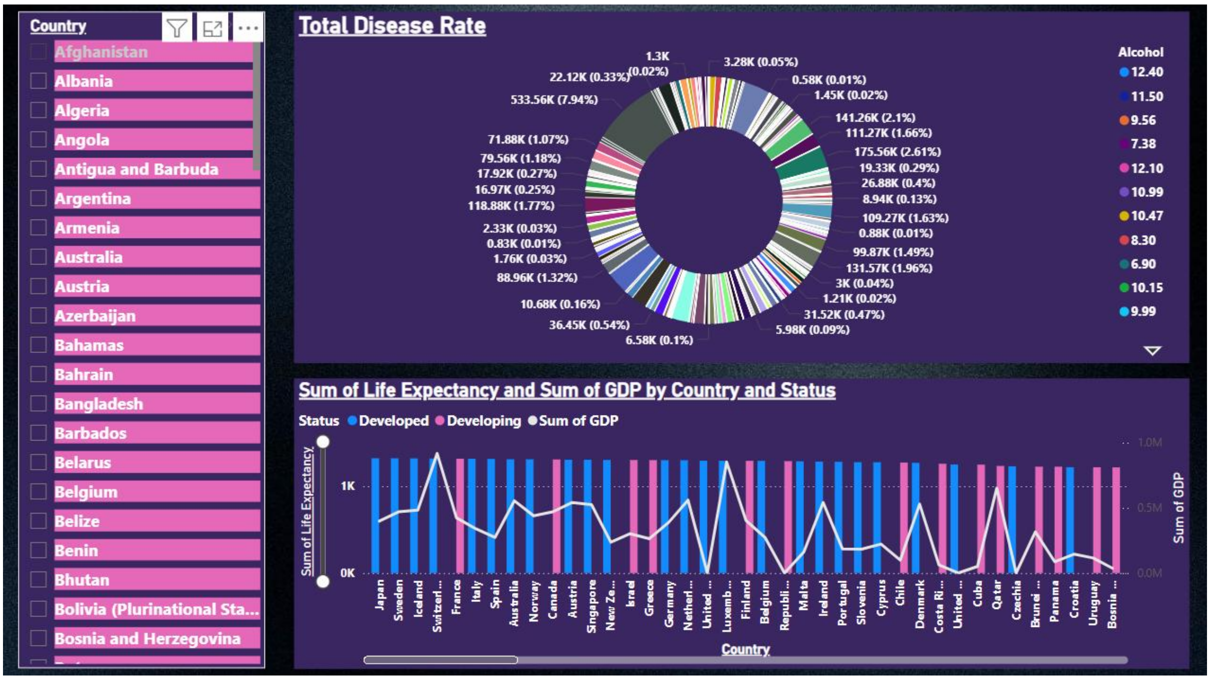
b) <u>Bar Chart – Adult Mortality by Country</u>

Displays adult mortality rates across countries. Helps identify countries with higher adult mortality rates.

How it was created:

In this visualization, the Country field is placed on the Axis to represent different countries, and Adult Mortality is added to the Values section to display the mortality rate for each country. This helps in comparing adult mortality levels across different countries in a simple and clear way.

DASHBOARD PAGE 3: DISEASE RATE ANALYSIS



a) <u>Country Slicer</u>

A vertical slicer listing all countries. Allows filtering of all visuals based on selected countries.

How it was created:

In this visualization, the Country field is added to a Slicer, which allows users to filter the data based on selected countries. This makes the dashboard interactive, as users can view and analyse information for specific countries as needed.

b) <u>Donut Chart – Total Disease Rate</u>

Shows the distribution of total disease rate across countries. Helps understand contribution of each country to overall disease rate.

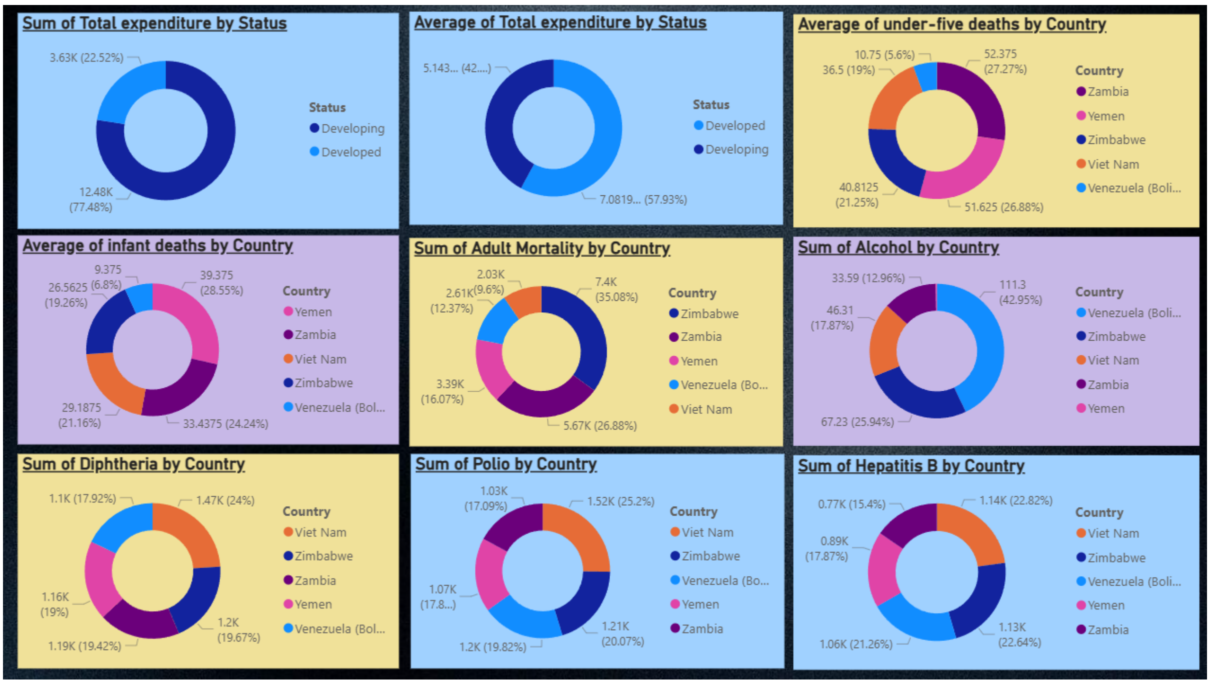
How it was created:
In this visualization, the Country field is placed in the Legend to represent different countries, while the disease-related measure is added to the Values section to show its magnitude. This helps in comparing how the disease-related values are distributed across different countries.

c) <u>**Combo Chart – Life Expectancy and GDP by Country and Status**</u>
A combination of bar and line chart showing:

- Life Expectancy (bars)
- GDP (line)
- Categorized by Developed/Developing

Shows relationship between economic condition (GDP) and life expectancy.
How it was created:
In this visualization, the Country field is placed on the Axis to represent different countries. Life Expectancy is added to the Column Values to display it as bars, while GDP is added to the Line Values to show it as a line. The Status field is used in the Legend to differentiate between developed and developing countries, helping to compare life expectancy and GDP across different country categories.

<u>DASHBOARD PAGE 4: OTHER HEALTH INDICATORS</u>



a) **Donut Charts (Multiple)**

Multiple donut charts are used for:

- Total Expenditure (Sum & Average)
- Under-five deaths
- Infant deaths
- Adult mortality
- Alcohol consumption
- Diseases like Diphtheria, Polio, Hepatitis B

These visuals show distribution and comparison of different health indicators across countries.

How they were created:

In this visualization, the Country field is placed in the Legend to represent different countries, while the corresponding metric (such as Alcohol consumption, Polio, etc.) is added to the Values section. This helps in showing how each country contributes to that particular health indicator and allows for easy comparison across countries.

7. Key Insights

Page1: Life Expectancy Analysis

Highest Performing: Austria, Germany, and the United Kingdom lead with the highest life expectancy at 86-88.
Lowest Performance: The lowest recorded is 49.70, which shows how the countries with the lowest performance (most probably in Sub-Saharan African countries) are underperforming.
Trend Over Time: The year slicer shows data from 2000 to 2015. The year chosen is 2012, which shows the high values of the top countries to be steady over the years, indicating that the life expectancy of developed countries plateaued at the 80s range.
Important Observation: The Min of BMI = 0.00 and the Min of Thinness = 0.00 indicate that these are almost certainly the result of the preprocessing stage where the missing data is replaced with zeros. This would probably affect the averages calculated (BMI avg 39.66, thinness avg 4.53) as well as the correlation analysis done with these fields.

Page 2: Infant & Adult Mortality

Highest Performing: The developed countries like the United States are concentrated around 0K on the y-axis, which indicates robust health infrastructures with negligible infant mortality rates.
Lowest Performance: India and Nigeria are way out in front with respect to infant deaths at ~20K+ which is significantly higher than all other countries on the chart. The gap between these two and the third country on this chart (Pakistan) is enormous and close to double.
Trend Over Time: Since this chart is a cumulative chart (Sum), countries with larger populations and longer data sets will naturally be higher on this chart. This makes it

difficult to analyse this chart with respect to year-over-year trends. Perhaps an average value would be more helpful.
Important Observation: In the case of Adult Mortality, Lesotho, Zimbabwe, and Botswana are at the top of this chart. All these countries are from Southern Africa and were significantly affected by HIV/AIDS.

Page 3: Disease Rate

Highest Performing: Nations like Japan, Sweden and Iceland have the highest Life Expectancy + GDP which reiterates the direct correlation between the wealth of a nation and the longevity of its people.
Lowest Performance: From the Total Disease Rate donut chart, it's evident that the highest single value contributes 533.56K cases (7.94%) of the total burden implying that a particular nation bears a disproportionately higher burden of disease compared to the rest.
Trend Over Time: From the combo chart, it's evident that the developed nations (high GDP) represented by the blue-coloured bars have a higher life expectancy at all times, whereas the life expectancy of the developing nations represented by the pink-coloured bars is lower, reiterating the direct correlation between GDP and life expectancy.
Important Observation: From the alcohol consumption legend placed on the right of the chart, ranging from 6.90 to 12.40, it's evident that the nations bearing a higher disease burden also have a higher alcohol consumption level.

Page 4: Other Reasons by Country

Highest Performing: Developed countries have an approximately 57.93% share in the average total health expenditure, whereas developing countries have only 42%. This indicates a greater investment in health for the developed world.
Lowest Performance: Zimbabwe is represented on the Adult Mortality, Alcohol, Diphtheria, Polio, and Hepatitis B charts. This indicates it is the country with the highest risk among all the health aspects on this page.
Trend Over Time: This page shows how different countries perform on different health aspects and how countries with lower health expenditures have more health problems over the years.
Important Observation: Five countries appear repeatedly on almost every chart on this page, which are Vietnam, Zimbabwe, Venezuela, Yemen, and Zambia. This indicates that these countries have critical problems with almost all health aspects at the same time and therefore need to be focused on more than any other countries on this dashboard.

8. Conclusion

This Power BI project analysed how socio-economic and health-related factors influence life expectancy across 183 countries from 2000 to 2015 using a Kaggle dataset. After preprocessing the data, four interactive dashboards were built to present the findings clearly.
The first dashboard established a global overview, revealing an average life expectancy of 70.92 years with developed nations like Austria, Germany and the United Kingdom

consistently leading with values in the high 80s, while some countries fell as low as 49.70 years.

The second dashboard highlighted the stark mortality divide between developed and developing nations with India and Nigeria recording the highest infant deaths and Southern African countries like Lesotho and Zimbabwe dominating adult mortality which is largely driven by the HIV/AIDS epidemic.

The third dashboard confirmed a strong positive correlation between GDP and life expectancy with wealthier nations consistently achieving better health outcomes while also identifying that a small number of countries carry a disproportionately large share of the global disease burden.

The fourth dashboard revealed that developed nations account for nearly 57.93% of total health expenditure and identified Vietnam, Zimbabwe, Venezuela, Yemen and Zambia as the most consistently vulnerable countries across multiple health indicators.

Overall, the project clearly demonstrated that life expectancy is shaped by an interconnected combination of economic strength, healthcare investment and disease burden with the greatest disparities falling along the developed-developing nation divide.